ESP32 GATE CONTROLLER — HARDWARE REFERENCE

Firmware development reference · All hardware defined · Rev 2.0

Relays   Reed switches   Buttons   Display / SPI   RF CC1101   TRIAC / Power   Power   ADC / Sensors   Aux relays

DIRECTION RELAYS (EXISTING)

L1 Open	IN1	GPIO 25
L1 Close	IN2	GPIO 26
L2 Open	IN3	GPIO 27
L2 Close	IN4	GPIO 14

Active HIGH. Optocoupler module. These relays switch direction only — never under load when TRIAC is active.

AUX RELAYS NEW

AUX 1 (lock)	IN5	GPIO 13
AUX 2 (light)	IN6	GPIO 12

Configurable. Default: AUX1 = electromagnetic lock (releases before opening, re-engages after closing). AUX2 = garage light / siren. Both expose NO/NC/COM on connector.

REED SWITCHES (EXISTING)

L1 Open	LOW = active	GPIO 34
L1 Close	LOW = active	GPIO 35
L2 Open	LOW = active	GPIO 32 ↑
L2 Close	LOW = active	GPIO 33 ↑

GPIO 34/35: input-only, no internal pull-up → 10kΩ external to 3.3V required. GPIO 32/33: INPUT_PULLUP enabled.

BUTTONS (EXISTING)

GATE toggle	INPUT_PULLUP	GPIO 22
MINUS [-]	INPUT_PULLUP	GPIO 16
PLUS [+]	INPUT_PULLUP	GPIO 17
OK (3s=WiFi rst)	INPUT_PULLUP	GPIO 0

All active LOW. GPIO 0 = boot pin — button

press during power-on enters bootloader. Keep debounce ≥50ms.

ESP32-WROOM · 38 pins

25	RELAY_L1_OPEN	OUT	26	RELAY_L1_CLOSE	OUT
27	RELAY_L2_OPEN	OUT	14	RELAY_L2_CLOSE	OUT
13	RELAY_AUX1	OUT	12	RELAY_AUX2	OUT
34	REED_OPEN_L1	IN⚠	35	REED_CLOSE_L1	IN⚠
32	REED_OPEN_L2	IN↑	33	REED_CLOSE_L2	IN↑
21	TRIAC_GATE	PWM	36	ZERO_CROSS	IN⚠
39	ACS712_OUT	ADC⚠	22	BTN_GATE	IN↑
16	BTN_MINUS	IN↑	17	BTN_PLUS	IN↑
0	BTN_OK	IN↑⚠	5	TFT_CS	SPI
2	TFT_DC	OUT	4	TFT_RST	OUT
23	SPI_MOSI	SPI	18	SPI_SCK	SPI
19	SPI_MISO	SPI	15	CC1101_CSN	SPI
3	CC1101_GDO0	IN	3V3	VCC 3.3V	PWR
GND	GND	PWR	VIN	5V (HLK)	PWR

⚠ GPIO CONSTRAINTS

GPIO 34,35,36,39	Input-only. No pull-up. No output.
GPIO 0	Boot pin. LOW on power-on = bootloader.
GPIO 2	Boot indicator LED on some devkits.
GPIO 3 (RX0)	Used for GDO0 — Serial disabled in production.
GPIO 12	Flash voltage strapping — avoid HIGH on boot.
SPI bus	TFT and CC1101 share MOSI/SCK. CS selects device.

TRIAC POWER STAGE NEW

TRIAC	BTA24-600	600V 25A
Gate opto	MOC3021	no ZC
ZC detect	4N25 opto	GPIO 36
Snubber	100Ω + 10nF	across T1-T2
Gate PWM	LEDC timer	GPIO 21

220VAC on T1/T2. Mandatory creepage ≥6mm from LV on PCB. MOC3021 provides galvanic isolation. Never connect TRIAC gate directly to ESP32.

Motor common wire (shared by L1 and L2) connects through TRIAC. Direction relays switch with TRIAC OFF (zero current on contacts).

CURRENT SENSOR NEW

IC	ACS712-20A	
Output	ADC GPIO 39	ADC1_CH3
Sensitivity	100 mV/A	
Vout @ 0A	VCC/2 = 1.65V	
Position	motor common	M1+M2 total

Measures combined current of both motors. For per-motor measurement, add second ACS712 on L2 common. GPIO 39 = ADC1 — use ADC1 only (ADC2 disabled when WiFi active).

CC1101 RF 433MHZ NEW

VCC	3.3V	
CSN	SPI CS	GPIO 15
SCK	SPI shared	GPIO 18
MOSI	SPI shared	GPIO 23
MISO	SPI shared	GPIO 19
GDO0	OOK data out	GPIO 3
Mode	OOK transparent	RC-switch compat

GDO0 configured as async serial data output in OOK mode. RC-switch library reads GPIO 3 via interrupt. SPI bus shared with TFT — TFT_CS (5) and CC1101_CSN (15) must never be LOW simultaneously.

ST7735 DISPLAY (EXISTING)

CS	SPI CS	GPIO 5
DC	Data/Cmd	GPIO 2
RST	Reset	GPIO 4
MOSI	SPI shared	GPIO 23
SCK	SPI shared	GPIO 18
MISO	not connected	—
Size	160×128 px	ST7735S

POWER ARCHITECTURE

AC input	220VAC (BR)
Surge protection	MOV varistor
AC fuse	2A slow-blow
PSU	HLK-5M05 → 5VDC
3.3V reg	AMS1117 (on ESP32 module)
Relay coils	5VDC
TRIAC supply	directly from 220VAC
Isolation	MOC3021 + 4N25

FIRMWARE FEATURES — STATUS

State machine	✓ done
Reed / sequence	✓ done
WiFi + Web UI	✓ done
NVS config + log	✓ done
TFT display menu	✓ done
Aux relays (lock/light)	⟳ next
RF CC1101 / RC-switch	⟳ next
Auto-close / pedestrian	⟳ next
TRIAC phase-cut	○ hw pending
ACS712 current sense	○ hw pending
HomeKit (HomeSpan)	○ later
MQTT	○ later

SUGGESTED FW SEQUENCE

Stage 7	Aux relays + lock logic
Stage 8	Auto-close + pedestrian mode
Stage 9	RF CC1101 + RC-switch + pairing
Stage 10	TRIAC phase-cut + soft-start
Stage 11	ACS712 + anti-crush + fusível virtual
Stage 12	HomeKit (HomeSpan)
Stage 13	MQTT + remote access
Stage 14	OTA firmware update

> Stages 10-11 require new hardware. Stages 7-9 run on current hardware.